

Unit 3A: Uses and Abuses of Percentages

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1. Foundation: What Are Percentages?

Definition

A percentage is a way of expressing a number as a fraction of 100. The word "percent" literally means "per hundred."

Key Insight: Percentages are always **relative** measures, not absolute ones.

Real-Life Foundation Example: Battery Life When your phone shows 75% battery, it means you have 75 out of every 100 units of battery capacity remaining. If your battery holds 3000 units total, you have 2,250 units left (75% of 3000).

Think About This: If I tell you that sales increased by 50%, what crucial information is missing?

2. Calculation Methods and Formulas

Basic Percentage Calculations

Finding a Percentage of a Number

Formula: $(\text{Percentage} \div 100) \times \text{Number}$ **Example:** What is 25% of 80?

- $(25 \div 100) \times 80 = 0.25 \times 80 = 20$

Finding What Percentage One Number Is of Another

Formula: $(\text{Part} \div \text{Whole}) \times 100$ **Example:** What percentage is 15 out of 60?

- $(15 \div 60) \times 100 = 0.25 \times 100 = 25\%$

Percentage Change

Formula: $((\text{New Value} - \text{Old Value}) \div \text{Old Value}) \times 100$ **Example:** Price increased from \$40 to \$50

- $((50 - 40) \div 40) \times 100 = (10 \div 40) \times 100 = 25\% \text{ increase}$

Critical Distinction: Percentage Points vs. Percentages

Scenario: Interest rate changes from 2% to 3%

- **Percentage increase:** $((3 - 2) \div 2) \times 100 = 50\% \text{ increase}$
- **Percentage point increase:** $3 - 2 = 1 \text{ percentage point increase}$

Real-Life Calculation Examples

Restaurant Tipping Scenario

Bill total: \$45.00 You want to tip 18%:

- Quick method: $\$45 \times 0.18 = \8.10
- Mental math method: 10% = \$4.50, so 18% = \$4.50 + \$3.60 = \$8.10

Salary Negotiation Example

Current salary: \$50,000 Offered raise: \$55,000 Percentage increase: $((\$55,000 - \$50,000) \div \$50,000) \times 100 = 10\%$

But here's where it gets interesting: If you were making \$50,000 and got promoted to a \$65,000 position, that's a 30% increase. The dollar difference (\$15,000) is the same whether you go from \$35,000 to \$50,000 (43% increase) or from \$50,000 to \$65,000 (30% increase).

Why This Matters: These describe very different magnitudes of change!

3. Legitimate Uses of Percentages

When Percentages Are Helpful

1. Comparing Different Scales

Example: Test Scores

- Class A: 18 out of 20 students passed = 90%
- Class B: 27 out of 30 students passed = 90%
- Both classes performed equally well despite different sizes

2. Tracking Changes Over Time

Example: Population Growth

- City A grew from 10,000 to 11,000 people = 10% growth
- City B grew from 100,000 to 110,000 people = 10% growth
- Same rate of growth, different absolute numbers

3. Financial Applications

- **Interest rates:** 5% annual return on investment
- **Discounts:** 20% off original price
- **Tax rates:** 15% income tax

4. Common Abuses and Misconceptions

Abuse #1: Missing Base Information

Misleading Claim: "Crime increased by 200% in our neighborhood!" **Hidden Truth:** Increased from 1 incident to 3 incidents **Why It's Misleading:** Sounds dramatic, but the absolute change is minimal

Lesson: Always ask "200% of what?"

Abuse #2: Cherry-Picking Time Periods

Example: Stock Performance Claims

- **Misleading:** "Our fund gained 50% this year!"
- **Full Picture:** Fund lost 60% last year, 30% the year before
- **Reality:** Still significantly down overall

Local News Drama: "Burglaries increase 300% in Maple Street neighborhood!" **Hidden truth:** Increased from 1 burglary last year to 4 this year **Why it's misleading:** Sounds like a crime wave, but it's just a few isolated incidents

Abuse #3: Inappropriate Comparisons

Problematic Example: "90% of students prefer our textbook over the competition" **Hidden Issues:**

- How many students were surveyed?
- How many textbooks were compared?
- What was the methodology?
- Were students given incentives?

Abuse #4: The Percentage Point Confusion

Media Headline: "Unemployment rate jumps 50%!" **Reality Check:**

- Rate went from 4% to 6%
- That's a 2 percentage point increase
- OR a 50% relative increase
- Very different implications!

Abuse #5: Meaningless Percentages

Black Friday Sale: "Everything 50% off! Plus members get an additional 20% discount!" **What shoppers think:** 70% total discount **Actual calculation:**

- Original price: \$100
- After 50% off: \$50
- After additional 20% off the sale price: $\$50 \times 0.8 = \40
- Total discount: 60%, not 70%

Silly Example: "Give 110% effort!" **Problem:** You cannot give more than 100% of something that represents totality

Legitimate Version: "Increase your effort by 110%" (meaning 2.1 times your previous effort)

5. Real-World Applications

Financial Context Examples

Credit Cards and Interest

Scenario: Credit card charges 24% APR

- **What this means:** If you owe \$1000 for a full year, you'll pay \$240 in interest
- **Monthly equivalent:** Approximately 2% per month
- **Compound effect:** Interest builds on previous interest

Sales and Discounts

Common Trick: "50% off everything! Plus an additional 20% off!"

- **Customer thinks:** 70% discount
- **Reality:** First 50% off, then 20% off the reduced price
- **Actual discount:** $50\% + (20\% \times 50\%) = 50\% + 10\% = 60\%$

Calculation:

- Original price: \$100
- After 50% off: \$50
- After additional 20% off: $\$50 \times 0.8 = \40
- Total discount: 60%, not 70%

Statistical Reporting

Survey Results

Proper Reporting: "In a survey of 1,000 randomly selected adults, 65% reported satisfaction with the service (margin of error $\pm 3\%$)"

Red Flags to Watch For:

- No sample size mentioned
- No methodology described
- No margin of error provided
- Loaded questions not disclosed

Medical and Scientific Claims

Good Example: "Drug X reduced symptoms in 80% of 500 patients in a double-blind, placebo-controlled study"

Suspicious Example: "Drug Y works for 9 out of 10 people!" (No context, no study details, no comparison group)

6. Critical Thinking Framework

Questions to Always Ask

1. **What's the base?** Percentage of what exactly?
2. **What's the time frame?** Over what period?
3. **What's being compared?** Is the comparison fair and relevant?
4. **What's the sample size?** How many data points?
5. **What's the methodology?** How was this measured or calculated?
6. **What's the context?** Are there other factors to consider?

Red Flag Phrases

- "Up to X%" (focuses on maximum, not typical)
- "As much as X%" (same issue as above)
- "X% more!" (more than what?)
- "Studies show X%" (which studies? how many?)
- "X times more effective" (more than what baseline?)

The "So What?" Test

Always ask: Even if this percentage is accurate, what does it actually mean in practical terms?

Example: "Our new process is 200% faster!"

- **If old process:** 3 hours → **New process:** 1 hour (good improvement)
- **If old process:** 3 seconds → **New process:** 1 second (negligible real-world difference)

Key Takeaways

1. **Percentages are powerful but can be manipulative** - always demand context
2. **The base matters more than the percentage** - 100% of nothing is still nothing
3. **Time frames are crucial** - short-term percentages can hide long-term trends
4. **Sample sizes and methodologies matter** - especially for survey percentages
5. **Multiple percentages can compound in unexpected ways** - don't just add them
6. **Critical thinking beats quick calculations** - understanding context trumps computational speed

Remember: The goal isn't to become suspicious of all percentages, but to become a more informed consumer of numerical information. Percentages are incredibly useful tools when used honestly and understood properly.